

THE RANKER'S MASTERBOOK

LIMITS.

The Edge of Calculus

The Problem Set

100

PROBLEMS

10

CHAPTERS

3–5

DIFFICULTY

lim

One hundred original limits where the obvious move is the wrong move – for the very top of JEE Advanced and the Mathematical Olympiad. Solutions in the companion volume.

Contents

Ten chapters, ten problems each.

I	The Greatest Integer Function · $\lfloor x \rfloor$	Nº1–10
II	The Fractional Part · $\{x\}$	Nº11–20
III	The Signum Function · sgn	Nº21–30
IV	Absolute Value & Corners · x	Nº31–40
V	Inverse Circular Functions · $\tan^{-1}$	Nº41–50
VI	Logarithm & Exponential · $\ln$	Nº51–60
VII	Advanced Trigonometry · $\sin$	Nº61–70
VIII	Exotic Indeterminate Forms · 1^∞	Nº71–80
IX	Sandwich & Discrete Sums · Σ	Nº81–90
X	The Grand Hybrids · ∞	Nº91–100

CHAPTER I

The Greatest Integer Function



Where the graph leaps, the limit hides.

№1 The Sinc Just Misses

●●●●● L4

$$\lim_{x \rightarrow 0^+} \left\lfloor \frac{100 \sin x}{x} \right\rfloor$$

floor

sinc

one-sided

$x \rightarrow 0^+$

squeeze

№2 Sine From Below, Tangent From Above

●●●●● L4

$$\lim_{x \rightarrow 0^+} \left(\left\lfloor \frac{50 \sin x}{x} \right\rfloor + \left\lceil \frac{50 \tan x}{x} \right\rceil \right)$$

floor

sinc

tanc

one-sided

$x \rightarrow 0^+$

№3 It Wanted To Be One

●●●●● L4

$$\lim_{x \rightarrow +\infty} \left\lfloor x \sin \frac{1}{x} \right\rfloor$$

floor

sinc

$x \rightarrow +\infty$

one-sided-from-below

№4 Half From The Triangle

●●●●● L3

$$\lim_{x \rightarrow +\infty} \frac{1}{x^2} \sum_{k=1}^{\lfloor x \rfloor} k$$

floor

summation

$x \rightarrow +\infty$

squeeze

№5 Curvature Catches The Floor

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} (1 - \cos x) \left\lfloor \frac{1}{x^2} \right\rfloor$$

floor

reciprocal-square

$x \rightarrow 0^+$

squeeze

cosine

№6 The Inner Half

● ● ● ● ● L4

$$\lim_{n \rightarrow +\infty} \left\{ \sqrt{n^2 + n} \right\} \quad (n \in \mathbb{Z}^+)$$

floor

fractional-part

conjugate

$x \rightarrow +\infty$

integers

№7 Two Floors Meet At An Integer

● ● ● ● ● L4

$$\lim_{x \rightarrow 2^-} \left(\lfloor x^2 \rfloor - \lfloor x \rfloor^2 \right)$$

floor

integer-point

$x \rightarrow 2^-$

jump

composition

№8 The Logarithm Forgives The Floor

● ● ● ● ● L3

$$\lim_{x \rightarrow +\infty} \left(\ln \lfloor e^x \rfloor - x \right)$$

floor

logarithm

exponential

$x \rightarrow +\infty$

squeeze

№9 The Removable Ceiling

● ● ● ● ● L3

$$\lim_{x \rightarrow 0} \lfloor \cos x \rfloor$$

floor

cosine

$x \rightarrow 0$

two-sided

value-vs-limit

№10 Mind The Sign Of Zero

● ● ● ● ● L3

$$\lim_{x \rightarrow 0^-} \frac{\lfloor x \rfloor}{x}$$

floor

one-sided

$x \rightarrow 0$

non-existence

comparison

CHAPTER II

The Fractional Part



The sawtooth that never rests.

№11 The Wrong Branch at Two

● ● ● ● ● L3

$$\lim_{x \rightarrow 2^-} \frac{\{x\} - 1}{x - 2}$$

floor one-sided $x \rightarrow 2^-$ jump removable-looking

№12 Approaching Zero From Below

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^-} (1 + \{x\})^{1/x}$$

floor one-sided $x \rightarrow 0^-$ exponential 1^∞ -trap

№13 Sawtooth Meets Taylor

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} \frac{\{\tan x\} - \{\sin x\}}{x^3}$$

fractional-part $x \rightarrow 0^+$ taylor tan sin cubic

№14 Bounded Sawtooth at Infinity

● ● ● ● ● L4

$$\lim_{n \rightarrow \infty} \{\sqrt{n^2 + n}\}$$

fractional-part $x \rightarrow \infty$ conjugate sqrt bounded-frac

№15 The Tamed Oscillator

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} x \left\{ \frac{1}{x} \right\}$$

fractional-part $x \rightarrow 0^+$ squeeze oscillation naive-DNE

№16 Euler Hidden in the Sawtooth

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left\{ \frac{n}{k} \right\}$$

fractional-part

n→infinity

riemann-sum

harmonic

euler-mascheroni

№17 Cosine Slips Below One

● ● ● ● ● L3

$$\lim_{x \rightarrow 0^+} (\{\sin x\} + \{\cos x\})$$

fractional-part

x→0+

one-sided

sin

cos

№18 Two Sawtooths Cancel

● ● ● ● ● L4

$$\lim_{x \rightarrow 1} (\{x\} + \{2 - x\})$$

fractional-part

x→1

removable

two-sided

complementary

№19 One Side Lives, One Side Explodes

● ● ● ● ● L4

$$\lim_{x \rightarrow 0} \frac{\{x\}}{x} \quad (\text{compare the two one-sided limits})$$

fractional-part

x→0

one-sided-comparison

non-existence

floor

№20 The Arctangent Plateau

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} \frac{\{\arctan \frac{1}{x}\} - (\frac{\pi}{2} - 1)}{x}$$

fractional-part

x→0+

arctan

floor

derivative-form

CHAPTER III

The Signum Function

sgn

A function with only three answers – and it changes its mind.

№21 Sign of a Vanishing Gap

●●●●● L4

$$\lim_{x \rightarrow 0^+} \operatorname{sgn}(x - \sin x) \cdot \frac{1 - \cos x}{x^2}$$

signum

one-sided

$x \rightarrow 0^+$

taylor

1-cos

№22 Sign Guarding a Difference of Squares

●●●●● L3

$$\lim_{x \rightarrow 2^-} \operatorname{sgn}(4 - x^2) \cdot \frac{\arctan(x - 2)}{x - 2}$$

signum

one-sided

$x \rightarrow 2^-$

arctan

factoring

№23 The Hidden Signum

●●●●● L5

$$\lim_{x \rightarrow 0^+} \frac{2}{\pi} \arctan\left(\frac{1}{x - \tan x}\right)$$

signum

hidden

arctan

tan

one-sided

$x \rightarrow 0^+$

composition

№24 Sign of the Tangent Excess

●●●●● L4

$$\lim_{x \rightarrow 0^+} \operatorname{sgn}(\tan x - \sin x) \cdot \frac{1 - \cos x}{x \tan x}$$

signum

$x \rightarrow 0^+$

one-sided

tan

sin

1-cos

№25 Signum at the Horizon

●●●●● L3

$$\lim_{x \rightarrow +\infty} \operatorname{sgn}\left(\sin \frac{1}{x}\right) \cdot \arctan x$$

signum

$x \rightarrow +\infty$

sin

arctan

composition

№26 Sign of the Sine Across an Even Integer

● ● ● ● ● L4

$$\lim_{x \rightarrow 2^-} \operatorname{sgn}(\sin(\pi x)) \cdot \frac{2^x - 4}{x - 2}$$

signum

$x \rightarrow 2^-$

one-sided

$\sin(\pi x)$

exponential

$\ln$

№27 Inverse-Trig Tie-Breaker

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} \operatorname{sgn}(\arcsin x - \arctan x) \cdot \frac{\arcsin x - \arctan x}{x^3}$$

signum

$x \rightarrow 0^+$

one-sided

$\arcsin$

$\arctan$

series

composition

№28 Signum at Negative Infinity

● ● ● ● ● L4

$$\lim_{x \rightarrow -\infty} \operatorname{sgn}(x) \cdot (\sqrt{x^2 + x} + x)$$

signum

$x \rightarrow -\infty$

conjugate

surd

$\operatorname{sgn}(x)$

№29 Sign of a Fourth-Order Remnant

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} \operatorname{sgn}\left(\cos x - 1 + \frac{x^2}{2}\right) \cdot \frac{x^4}{\cos x - 1 + \frac{x^2}{2}}$$

signum

$x \rightarrow 0^+$

one-sided

cosine

fourth-order

series

№30 Floor Trap Beside the Signum

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} (\operatorname{sgn}(x) + \lfloor \cos x \rfloor) \cdot \frac{e^x - 1}{x}$$

signum

floor

$x \rightarrow 0^+$

one-sided

exponential

composition

CHAPTER IV

Absolute Value & Corners



Every corner is a trap. Split the limit before it splits you.

№31 The Cube That Forgives the Kink

●●●●● L4

$$\lim_{x \rightarrow 0} \frac{|x| - |\sin x|}{|x|^3}$$

modulus

sin

taylor

$x \rightarrow 0$

two-sided

cube

№32 A Factor Hidden in the Bars

●●●●● L3

$$\lim_{x \rightarrow 2} \frac{|x - 2| + |x^2 - 4|}{|x - 2|}$$

modulus

factoring

$x \rightarrow 2$

corner

two-sided

№33 When Sine and Tangent Both Bend

●●●●● L5

$$\lim_{x \rightarrow 0} \frac{|\sin x| - |x|}{|\tan x| - |x|}$$

modulus

sin

tan

$x \rightarrow 0$

two-sided

cancellation

№34 The Floor Tames the Modulus

●●●●● L4

$$\lim_{x \rightarrow 3} |x - 3| \left\lfloor \frac{1}{|x - 3|} \right\rfloor$$

modulus

floor

$x \rightarrow 3$

squeeze

two-sided

№35 Modulus at Negative Infinity

●●●●● L4

$$\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 + x} - |x| \right)$$

modulus

infinity

$x \rightarrow -\infty$

conjugate

sqrt

№36 The Self-Cancelling Inner Modulus

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} \frac{|x + |x||}{x}$$

modulus nested $x \rightarrow 0$ one-sided dne

№37 The Reciprocal Corner of Arctangent

● ● ● ● ● L4

$$\lim_{x \rightarrow 0} \frac{\arctan \frac{1}{|x|} - \frac{\pi}{2}}{|x|}$$

modulus arctan $x \rightarrow 0$ two-sided derivative-as-limit

№38 A Corner Raised to a Vanishing Power

● ● ● ● ● L5

$$\lim_{x \rightarrow 0} (|\cos x| + |x|)^{1/|x|}$$

modulus cos exponent $x \rightarrow 0$ 1^∞ log

№39 Two Nested Bars, One Clean Slope

● ● ● ● ● L3

$$\lim_{x \rightarrow 0} \frac{||x| - x| - ||x| + x|}{x}$$

modulus nested $x \rightarrow 0$ two-sided signum

№40 The Corner of Sine at Pi

● ● ● ● ● L5

$$\lim_{x \rightarrow \pi} \frac{|\sin x| - |x - \pi|}{|x - \pi|^3}$$

modulus sin $x \rightarrow \pi$ taylor two-sided corner

CHAPTER V

Inverse Circular Functions

$\tan^{-1}$

Branch cuts, boundary singularities, the $\pm\pi/2$ cliff.

№41 The Reciprocal Reflection

●●●●● L4

$$\lim_{x \rightarrow 0^+} \frac{\arctan\left(\frac{1}{x}\right) - \frac{\pi}{2}}{x}$$

arctan

reciprocal

$x \rightarrow 0^+$

asymptotic

one-sided

№42 Twin Telescopes at Infinity

●●●●● L4

$$\lim_{x \rightarrow +\infty} x^2 \left(\arctan \frac{x+2}{x+1} - \arctan \frac{x+1}{x} \right)$$

arctan

subtraction-formula

$x \rightarrow \text{infinity}$

rational-argument

asymptotic

№43 The Vanishing Square Root

●●●●● L4

$$\lim_{x \rightarrow 1^-} \frac{\frac{\pi}{2} - \arcsin x}{\sqrt{1-x^2}}$$

arcsin

square-root-singularity

$x \rightarrow 1^-$

conjugate

endpoint

№44 Second Difference of the Arc

●●●●● L5

$$\lim_{x \rightarrow +\infty} x^3 \left(\arctan(x+1) + \arctan(x-1) - 2 \arctan x \right)$$

arctan

second-difference

$x \rightarrow \text{infinity}$

subtraction-formula

convexity

№45 Across the Branch Cut

● ● ● ● ● L5

$$\lim_{x \rightarrow (\pi/2)^+} \frac{\arctan(\tan x) + \frac{\pi}{2}}{x - \frac{\pi}{2}}$$

arctan tan branch-cut $x \rightarrow \pi/2$ one-sided nonexistence

№46 Crashing into Minus One

● ● ● ● ● L4

$$\lim_{x \rightarrow -1^+} \frac{\arcsin x + \frac{\pi}{2}}{\sqrt{x+1}}$$

arcsin endpoint $x \rightarrow -1^+$ square-root-singularity one-sided

№47 The Other End of Arctan

● ● ● ● ● L4

$$\lim_{x \rightarrow -\infty} x \left(\frac{\pi}{2} + \arctan x \right)$$

arctan $x \rightarrow -\infty$ reflection negative-infinity asymptotic

№48 Arcsine Raised to a Power

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} \left(\frac{\arcsin x}{x} \right)^{1/x^2}$$

arcsin 1-power-infinity log-transform series $x \rightarrow 0^+$

№49 Two Arctans, One Argument

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} \frac{\arctan(2x) - 2 \arctan x}{x^3}$$

arctan maclaurin-series $x \rightarrow 0^+$ cubic-cancellation odd-function

№50 The Floor Beneath the Arc

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} \arctan \left(\left\lfloor \frac{1}{x} \right\rfloor x \right)$$

arctan floor squeeze $x \rightarrow 0^+$ fractional-part

CHAPTER VI

Logarithm & Exponential

ln

Growth races, hidden orders, and the 1^∞ only logarithms can tame.

№51 The Logarithm's Echo

●●●●● L4

$$\lim_{x \rightarrow 0} \left(\frac{\ln(1+x)}{x} \right)^{1/x}$$

1^∞ infinity

$\ln(1+x)$

second-order

$x \rightarrow 0$

№52 Chasing Euler's Number

●●●●● L5

$$\lim_{x \rightarrow \infty} \left(\frac{\left(1 + \frac{1}{x}\right)^x}{e} \right)^x$$

1^∞ infinity

e-expansion

second-order

$x \rightarrow \infty$

№53 The Point Named e

●●●●● L4

$$\lim_{x \rightarrow e} (\ln x)^{\frac{1}{x-e}}$$

1^∞ infinity

$\ln$ -of- $\ln$

$x \rightarrow e$

nonzero-point

№54 The Buried Term

●●●●● L4

$$\lim_{x \rightarrow -\infty} e^{-x} \left(x - \ln(e^x + e^{2x} + e^{3x}) \right)$$

$\log$ -of-sum

dominated-term

$x \rightarrow -\infty$

exponential-tail

№55 Three Floors Down

● ● ● ● ● L5

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e + \frac{e x}{2}}{x^2}$$

x^x

third-order

$(1+x)^{1/x}$

$x \rightarrow 0$

brutal-expansion

№56 Change of Base, Change of Heart

● ● ● ● ● L3

$$\lim_{x \rightarrow 1} \left(\log_x e - \frac{1}{x-1} \right)$$

base-change

$\log_x$

$x \rightarrow 1$

infinity-minus-infinity

№57 Sine Under the Log

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} \frac{\ln(\sin x) - \ln x}{x^2}$$

log-of-ratio

one-sided

$x \rightarrow 0^+$

$\sin x$ -over- x

№58 The Slowly Widening Log

● ● ● ● ● L5

$$\lim_{x \rightarrow \infty} \left(\frac{\ln(x+1)}{\ln x} \right)^{x \ln x}$$

1^∞

base-change

$x \rightarrow \infty$

$\ln(x+1)$ -over- $\ln x$

№59 A Tale of Two Sides

● ● ● ● ● L4

$$\lim_{x \rightarrow 0} (1 + |x|)^{1/x}$$

nonexistence

one-sided

absolute-value

1^∞

$x \rightarrow 0$

№60 The Root That Refuses to Settle

● ● ● ● ● L5

$$\lim_{x \rightarrow \infty} \frac{x(x^{1/x} - 1) - \ln x}{(\ln x)^2/x}$$

$x^{1/x}$

second-order

$x \rightarrow \infty$

exp-of-log-over- x

CHAPTER VII

Advanced Trigonometry

sin

Beyond $\sin x$ – nested angles, telescopes, Taylor cancellations.

№61 The Iterated Sine Crawls Home

●●●●● L4

$$\text{Let } x_1 \in (0, \pi) \text{ and } x_{n+1} = \sin(x_n). \quad \lim_{n \rightarrow \infty} n x_n^2$$

iterated-sine

recursion

cesaro-stolz

asymptotics

x→infinity

№62 Nested Tangent Beats Nested Sine

●●●●● L4

$$\lim_{x \rightarrow 0} \frac{\tan(\tan x) - \sin(\sin x)}{\tan x - \sin x}$$

nested-trig

tan-tan

sin-sin

taylor

x→0

ratio

№63 A Pisot Number Hugs the Integers

●●●●● L5

$$\lim_{n \rightarrow \infty} (2 + \sqrt{3})^n \sin(\pi (2 + \sqrt{3})^n)$$

sin(pi*f)

near-integer

pisot

conjugate

x→infinity

sequence

№64 The Seventh-Order Duel

●●●●● L5

$$\lim_{x \rightarrow 0} \frac{\sin(\tan x) - \tan(\sin x)}{x^7}$$

sin-tan

tan-sin

high-order

taylor

x→0

cancellation

№65 Viète's Radical Telescope

●●●●● L4

$$\text{Let } a_n = \sqrt{2 + \underbrace{\sqrt{2 + \cdots + \sqrt{2}}}_{n \text{ twos}}}. \quad \lim_{n \rightarrow \infty} 2^n \sqrt{2 - a_n}$$

half-angle

nested-radical

telescope

viète

x→infinity

sequence

№66 A Power Tower on the Sinc

● ● ● ● ● L4

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{1-\cos x}}$$

one-to-infinity

log-transform

sinc

second-order

$x \rightarrow 0$

№67 The Floor That Drops to Zero

● ● ● ● ● L3

$$\lim_{x \rightarrow 0^+} \left\lfloor x \cot x \right\rfloor$$

floor

one-sided

$x \rightarrow 0^+$

cot

squeeze

№68 Three Cosines in Lockstep

● ● ● ● ● L3

$$\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x \cos 3x}{x^2}$$

product

cosine

second-order

$x \rightarrow 0$

weighted-sum

№69 Sine at the Edge of Minus Infinity

● ● ● ● ● L4

$$\lim_{x \rightarrow -\infty} x^3 \left(\sin \frac{1}{x} - \frac{1}{x} \right)$$

$x \rightarrow -\infty$

substitution

$\sin(1/x)$

third-order

cubic-term

№70 When Inverses Flip the Sign

● ● ● ● ● L5

$$\lim_{x \rightarrow 0} \frac{\tan(\sin x) - \sin(\tan x)}{\arcsin(\arctan x) - \arctan(\arcsin x)}$$

high-order

inverse-trig

arcsin-arctan

$x \rightarrow 0$

ratio

symmetry

CHAPTER VIII

Exotic Indeterminate Forms

1^∞

0^0 , $\infty - \infty$, ∞^0 – forms that hide their answer until forced.

№71 The Arctan Mirage

● ● ● ● ● L4

$$\lim_{x \rightarrow 0} \left(\frac{\arctan x}{x} \right)^{1/x^2}$$

1^∞

arctan

$x \rightarrow 0$

taylor

log-transform

№72 Vanishing Tail at Infinity

● ● ● ● ● L4

$$\lim_{x \rightarrow +\infty} \left(\frac{\pi}{2} - \arctan x \right)^{1/\ln x}$$

0^0

arctan

$x \rightarrow +\infty$

log-transform

asymptotics

№73 The Floor That Forgives

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} (1 + x)^{[1/x]}$$

1^∞

floor

$x \rightarrow 0^+$

fractional-part

squeeze

№74 The Logarithm's Second Secret

● ● ● ● ● L4

$$\lim_{x \rightarrow +\infty} \left(x - x^2 \ln \left(1 + \frac{1}{x} \right) \right)$$

infinity-minus-infinity

ln

$x \rightarrow +\infty$

taylor

collapse

№75 Fractional Part Meets Itself

● ● ● ● ● L3

$$\lim_{x \rightarrow 1^+} \{x\}^{\{x\}}$$

0^0

fractional-part

$x \rightarrow 1^+$

log-transform

№76 The Square-Root Disguise

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} (\cos \sqrt{x})^{1/x}$$

1^infinity

square-root

x→0+

taylor

half-integer-powers

№77 Modulus at the Far Left

● ● ● ● ● L4

$$\lim_{x \rightarrow -\infty} \left(\frac{|x| + 1}{|x| - 1} \right)^x$$

1^infinity

modulus

x→-infinity

asymptotics

log-transform

№78 Floor Times a Vanishing Arc

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} (\lfloor 1/x \rfloor \cdot \arcsin x)$$

0-times-infinity

floor

arcsin

x→0+

squeeze

№79 Signum Splits the Exponential

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} \left(1 + \operatorname{sgn}(x) x^2 \right)^{1/x^2}$$

1^infinity

signum

x→0+

one-sided

log-transform

№80 The Limit That Refuses

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} \left(x \lfloor 1/x \rfloor \right)^{1/x}$$

1^infinity

floor

fractional-part

x→0+

non-existence

oscillation

CHAPTER IX

Sandwich & Discrete Sums



When termwise intuition lies, only a global bound tells the truth.

№81 The Floor That Doesn't Bite

● ● ● ● ● L3

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n \left\lfloor k \frac{n}{k+1} \right\rfloor$$

floor

riemann-sum

x→infinity

integer-part-sum

№82 A Thousand Reciprocal Roots

● ● ● ● ● L4

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}}$$

1/sqrt(n^2+k)

squeeze

x→infinity

riemann-vs-squeeze

№83 Fractional Parts on a Grid

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left\{ \sqrt{k} \right\} \cdot \frac{1}{\sqrt{k}} \cdot \sqrt{n}$$

fractional-part

equidistribution

riemann-sum

x→infinity

№84 The Signum Mirage

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} x \sum_{k=1}^{\lfloor 1/x \rfloor} \operatorname{sgn} \left(\sin \frac{1}{kx} \right)$$

signum

sin

one-sided

x→0+

riemann-sum

squeeze

№85 Stolz Refuses the Easy Way

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} \frac{1}{(\ln n)^2} \sum_{k=1}^n \frac{\lfloor \log_2 k \rfloor}{k}$$

stolz-cesaro

floor

ln

x→infinity

dyadic

№86 Riemann With a Floor Inside

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left\{ \frac{n}{k} \right\}$$

floor

riemann-sum

x→infinity

equidistribution

№87 Conjugate Across the Whole Sum

● ● ● ● ● L4

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\sqrt{n^2 + k} - \sqrt{n^2 + k - 1} \right) \cdot \sqrt{\frac{k}{n}}$$

sqrt

global-bound

x→infinity

telescoping-fail

№88 Floor of an Arctan Riemann Sum

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n \left\lfloor n \arctan \frac{k}{n} \right\rfloor$$

floor

arctan

riemann-sum

x→infinity

global-squeeze

№89 The Misleading Termwise Limit

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{n^2} \sin^2 \left(\frac{k\pi}{n} \right)$$

termwise-trap

sin

global-bound

x→infinity

squeeze

№90 One-Sided Squeeze at an Integer

● ● ● ● ● L3

$$\lim_{x \rightarrow 2^-} (x - \lfloor x \rfloor) \left\lfloor \frac{1}{2 - x} \right\rfloor \cdot (2 - x)$$

floor

one-sided

x→2-

squeeze

CHAPTER X

The Grand Hybrids



Three or more worlds fused into a single pitiless limit.

№91 The Floor at the Threshold

● ● ● ● ● L4

$$\lim_{x \rightarrow 0^+} \left\lfloor \frac{1000 \sin x}{x} \right\rfloor$$

floor trigonometry one-sided $x \rightarrow 0^+$ squeeze

№92 Taming the Blow-Up

● ● ● ● ● L4

$$\lim_{x \rightarrow (\pi/2)^-} \left(\frac{\pi}{2} - x \right) \lfloor \tan x \rfloor$$

floor tangent one-sided blow-up squeeze

№93 A Tale of Two Cubics

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} \left| \frac{x - \sin x}{x - \arcsin x} \right|$$

floor sine arcsine series $x \rightarrow 0^+$

№94 The Fractional Mirage

● ● ● ● ● L5

$$\lim_{x \rightarrow 1^+} \left\{ \frac{\ln x}{x - 1} \right\}$$

fractional-part logarithm one-sided $x \rightarrow 1^+$ discontinuity

№95 The Vanishing Negative

● ● ● ● ● L4

$$\lim_{x \rightarrow -\infty} \lfloor x (\arctan x - \arctan(x - 1)) \rfloor$$

floor arctangent $x \rightarrow -\infty$ 0-times-bounded sign-of-zero

№96 Signum in the Exponent

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^-} \left(\frac{\arcsin |x|}{|x|} \right)^{\operatorname{sgn}(x)/x^2}$$

signum

arcsine

modulus

power

one-sided

log-transform

№97 Absolute Tangent

● ● ● ● ● L4

$$\lim_{x \rightarrow 0} \frac{|\tan x| - |x|}{|x|^3}$$

modulus

tangent

two-sided

series

$x \rightarrow 0$

№98 The Arctangent's Last Digit

● ● ● ● ● L5

$$\lim_{x \rightarrow +\infty} \left\lfloor -3x^3 \left(\frac{\pi}{2} - \arctan x - \frac{1}{x} \right) \right\rfloor$$

floor

arctangent

asymptotic

$x \rightarrow +\infty$

series

№99 Fracture at e

● ● ● ● ● L5

$$\lim_{x \rightarrow e^+} \frac{\{\ln x\}}{x - e}$$

fractional-part

logarithm

one-sided

$x \rightarrow e^+$

derivative

№100 The Capstone Squeeze

● ● ● ● ● L5

$$\lim_{x \rightarrow 0} \arctan \left(1 - |x| \left\{ \frac{1}{|x|} \right\} \right)$$

arctangent

fractional-part

modulus

floor

squeeze

$x \rightarrow 0$